Qianyu Zheng

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EDUCATION

Georgia Institute of Technology, Atlanta, GA *August 2022 – May 2026 (Expected)*

- Candidate for Bachelor of Science in Computer Science
- Year: Sophomore, GPA: 4.0

Skills

- Programming Languages: Python, Java, C, JavaScript
- Computer Science Knowledge: Machine Learning, Deep Learning, Databases, Graph Neural Networks, Computer Vision
- Concepts: Statistics, Linear Algebra, Combinatorics

RESEARCH

Georgia Tech Undergraduate Research, *Researcher* *May 2023 - Present*

- Researching the applications of Graph Neural Networks (GNNs) in material science, under the guidance of **Assoc. Prof. Victor Fung** at Georgia Tech.
- Played an integral role in integrating **contrastive learning** techniques into the pretraining process of GNNs to gain performance improvements within the CGCNN model under our framework "**matdeeplearn**."
- Conducted independent investigations to evaluate and improve **force field predictions of models** by applying **physically informed Machine Learning**. Implement physical models in matdeeplearn framework and design chemically meaningful evaluation metrics for ML models.
- Developed proficiency in Graph Neural Networks (GNNs), PyTorch framework, deep learning, and research methods.

PROJECTS

Participant, **Major-League Hacking Hacklytics 2023: Data Science Hackathon** *February 2023*

- Collaborated in a 4-member team to develop "Stock Tweet," a project tailored for **PR teams** of companies.
- Conducted analysis on potential impacts of sample tweets on stock prices and visually presented findings through an aesthetically pleasing word cloud.
- Applied **spaCy** package to processing retrieved tweets and trained a classification network on processed tweets to classify whether a tweet will positively or negatively influence the stock price of the company.
- Established a **MongoDB** database for storing data obtained from the Twitter API and engaging in close collaboration with team members to leverage the **gpt-2 model** for generating sample tweets based on trending keywords.

Team Leader, **Major-League Hacking HackGT 2023** *October 2023*

- **Led** a 4-member team in the development of "Plot Visualizer," a project with the objective of enhancing the **accessibility of STEM education** for students with learning difficulties by transforming graph data into data series through visualization. The project is still maintained and developed after the hackathon ends.
- Created a comprehensive dataset and implemented a fine-tuned **YOLOv7 model for object detection** and a fine-tuned **EasyOCR model for text recognition**, capable of handling challenging scenarios like blurred or tilted images with noise.
- Utilized OpenCV for image editing and visualization tasks.
- Developed a user-focused website using **Flask** for the backend and **HTML/CSS** for the frontend to showcase the entire data visualization pipeline.

ACTIVITIES

Backend Developer, **Georgia Tech Chinese Students and Scholars Association (GT CSSA)** *September 2022 – Present*

- Engage with GT CSSA, a collaborative organization uniting Chinese students to orchestrate campus-wide events and initiatives.
- Leveraged **Node.js** and **MongoDB** to engineer dynamic features, enabling password modifications via a token verification procedure.

Teaching assistant, **CS 1331: Introduction to Object Oriented Programming** *January 2023 – Present*

- Collaborate within a team of 42 TAs, led by Professor Richard Landry from the College of Computing.
- Conduct weekly 75-minute recitation sessions, engaging around 50 students. Acknowledged by students for providing consistently delivering clear explanations and enhancing their comprehension and academic performance. Honed **interpersonal communication skills**.
- Work as the **forum lead** that monitors the Q&A forum to actively answer questions from students and maintain good communication environment on the forum.
- Demonstrated proficiency in providing meticulous and tailored feedback to support student growth.